



# Furkan Karakas

Software and Quality Assurance Engineer

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## Summary

Software and quality assurance engineer with 5+ years of experience in software testing, test automation, developing full-stack applications, data integration solutions. Strong background in Java, Python, React.js, SQL/databases and test automation frameworks such as Selenium, Playwright and Appium. Adept at cloud technologies like Docker and CI/CD pipelines. Earned MSc. in Communication Systems from EPFL with 5.20/6.0 GPA.

## Technologies

**Programming:** Java, UI Automation (Appium, Selenium, Playwright), Python, TypeScript (React.js), Bash, C++

**Databases:** PostgreSQL, OracleDB, Data Warehousing

**Cloud/DevOps:** Docker, GitLab CI/CD, Talend (ETL)

**Languages:** English: C1, German: C1, Turkish: Native

**Other:** JIRA, HPQC, Graph Neural Networks, VaR Computation, Time Series Analysis

## Experience

**PostFinance AG**, Software Test Automation Engineer

Bern, Switzerland  
May 2024 – present  
2 years 4 months

- Automated independent end-to-end tests of the company web and mobile applications on the system level by means of black-box testing using an internal test automation toolbox built on top of Appium and Selenium
- Detected and reported countless bugs and issues in the diverse frontend and backend systems of the company, contributing to the overall quality of the applications
- Worked in four releases in a year using agile methodology
- Used the issue tracking service JIRA and the application lifecycle management tool HPQC

**ZZ Vermögensberatung (Schweiz) AG**, Financial Software Engineer

Vitznau, Switzerland  
July 2022 – Apr 2024  
1 year 10 months

- Developed internal "Knowledge Base" web app using React.js, Spring Boot, and PostgreSQL, streamlining business processes for 25+ colleagues
- Built automated CI/CD pipelines using GitLab runners in Docker for seamless application deployment to prod/dev environments, over 80% improvement of build time of the web application
- Created in-house Value-at-Risk computation framework in Python integrated with company data warehouse using Pandas/NumPy
- Parsed XML files to identify data load errors in Java, enabling team to review and correct issues daily

**Hoffmann-La Roche**, Master's Thesis Student

Basel, Switzerland  
Sept 2021 – Feb 2022  
6 months

- Conducted literature review on corporate skills management using machine learning
- Implemented and evaluated graph neural network method to learn skill representations
- Achieved 72% F1-score on skill prediction task by collaborating cross-functionally

**ABB Corporate Research Center**, Student Internship

- Designed Docker containers interacting on Hyperledger Fabric blockchain to store IoT sensor data
- Built React.js front-end application interfacing with the blockchain network

Baden, Switzerland

July 2020 – Dec 2020

6 months

## Education

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**Msc. Swiss Federal Institute of Technology Lausanne (EPFL)**, Computer and Communication Sciences

Lausanne, Switzerland

Sept 2019 – Apr 2022

- GPA: 5.20/6.00
- **Graduation thesis:** Graph Convolutional Networks for Skills Management in Large Organizations
- **Coursework:** Advanced probability and applications, Cryptography and security, Distributed algorithms, TCP/IP networking, Deep learning, Information security and privacy

**Bsc. Boğaziçi University**, Electrical Engineering & Mathematics

Istanbul, Türkiye

Feb 2015 – July 2019

- GPA: 3.59/4.00
- Admitted to Boğaziçi University (BOUN) by ranking 300th among 2.5 million candidates (**top 0.012%**) in the national entrance exam
- Graduated with double major in maths and electrical engineering with high honors
- **Coursework:** Pure mathematics and electrical engineering courses

**Istanbul High School**, High School Diploma

Istanbul, Türkiye

Sept 2009 – Aug 2014

- Admitted through high-ranking national exam performance
- Completed a rigorous dual-curriculum education primarily taught in German
- Awarded both the Turkish high school diploma and the Abitur, granting direct admission to universities in Germany, Switzerland, and worldwide